Synthetic Proton/Neutron Reconstruction Study

B = 1.15 T, Beam Deflection = 3.0°

SMSimulator5.5 Analysis

February 24, 2026

Study Goal and Configuration

Goal: Evaluate reconstruction quality with synthetic beam input and focus on

$$\Delta P_x = P_x^{\mathrm{reco}} - P_x^{\mathrm{true}}.$$

Configuration

- Magnetic setting from macro: `run_protons_B115T_3.0deg.mac`
- Geometry/field macro: `geometry_B115T.mac`
- Target from macro: $(-12.488, 0.009, -1069.458)\,\mathrm{mm}$, angle 3.0°
- Dipole field map: `180703-1,15T-3000.table`, rotation 30°

Synthetic Input Dataset

Generator setup

- Proton momentum: $(x_p, 0, 627)\,\mathrm{MeV}/c$, $x_p \sim \mathcal{U}[-150, 150]$
- Neutron momentum: $(x_n, 0, 627)\,\mathrm{MeV}/c$, $x_n \sim \mathcal{U}[-150, 150]$
- Generated events: 20,000
- Reconstruction analysis sample: 3,000 events

Event statistics in analyzed sample

- Events with reconstructed PDC track: 2597 / 3000 ($\sim$ 86.6%)
- Events with neutron TOF reconstruction: 994 / 3000 ($\sim$ 33.1%)
- Proton Minuit attempts: 600 events, Minuit success: 344 ($\sim$ 57.3%)

Reconstruction Pipelines

Proton (charged, PDC-based)

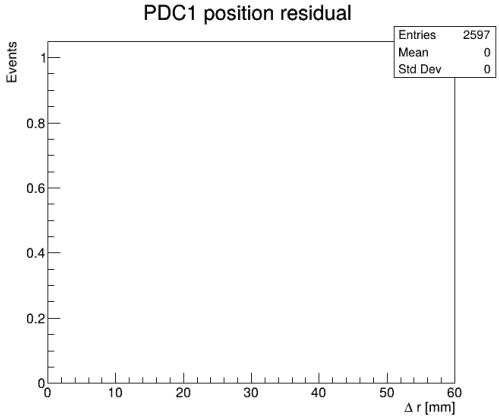
- Build PDC track from PDC1/PDC2 reconstructed points.
- Method A: **TMinuit** back-propagation in field to target.
- Method B: **Three-point constrained** fit (target + two PDC points).

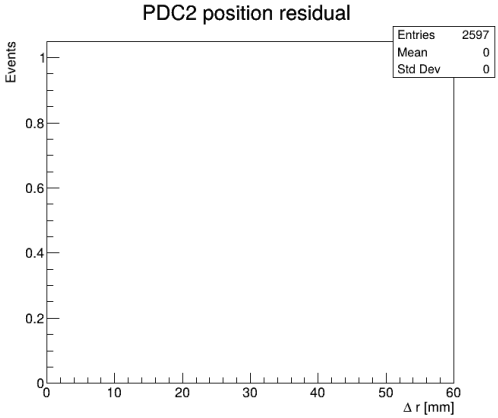
Neutron (neutral, NEBULA TOF)

- Cluster NEBULA hits in time window.
- Reconstruct hit position and TOF.
- Start point fixed at target (from macro), direction = target → hit.
- Compute β, γ, and 3D momentum:

$$\beta = \frac{L/t}{c}, \quad p = \gamma m_n \beta c.$$

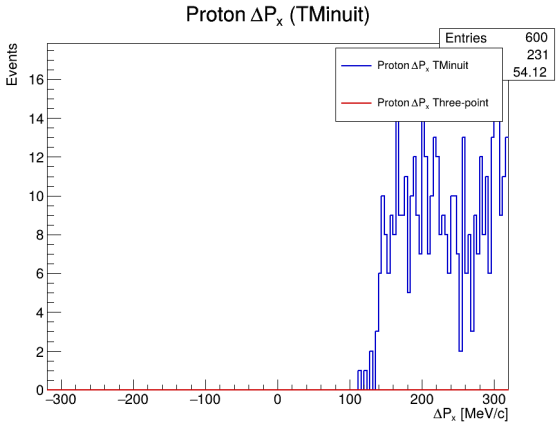
PDC Position Precision

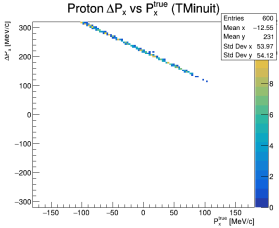


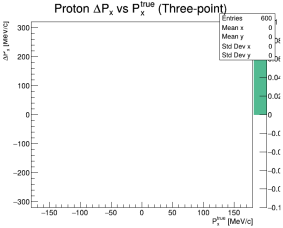


In this synthetic setup, PDC residual histograms peak at 0 mm in the analyzed events ($\langle \Delta r_{PDC1} \rangle = 0$, $\langle \Delta r_{PDC2} \rangle = 0$).

Proton ΔP_x: Key Result

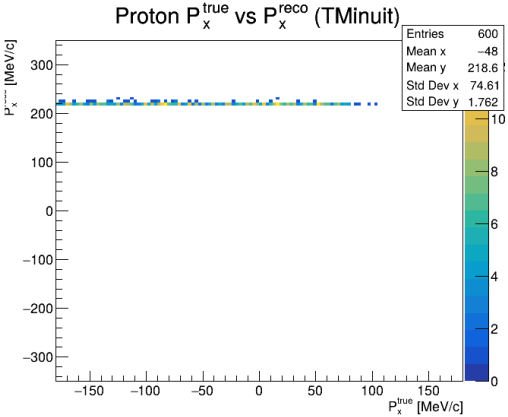


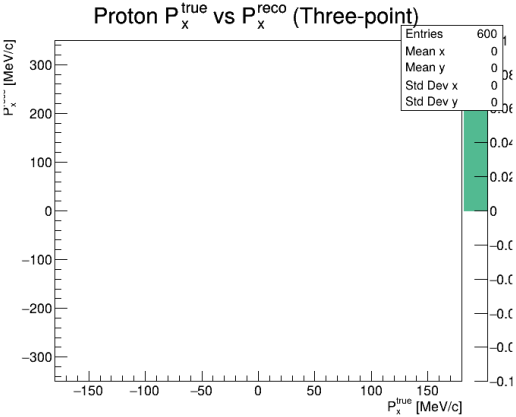




Method	Count	Mean(ΔP_x)	RMS(ΔP_x)
TMinuit	600	+230.969 MeV/c	54.115 MeV/c
Three-point	600	0.000 MeV/c	0.000 MeV/c

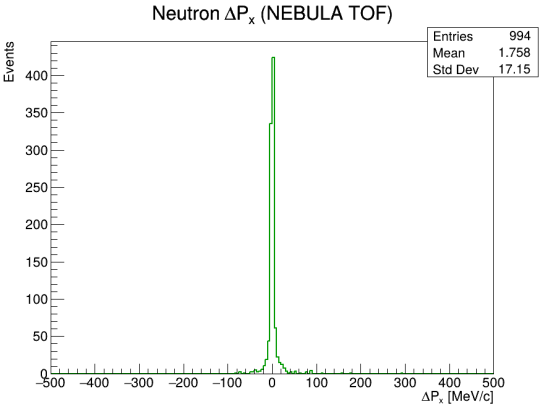
Proton P_x^{true} vs P_x^{reco}

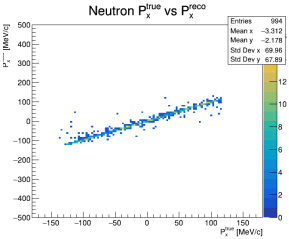


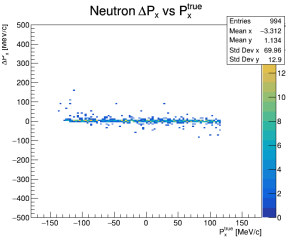


The Minuit branch shows a clear positive P_x bias in this sample.

Neutron ΔP_x and 3D Momentum







Neutron TOF reconstruction summary (994 events):

$$\text{Mean}(\Delta P_x) = +1.758\ \mathrm{MeV}/c, \quad \text{RMS}(\Delta P_x) = 17.152\ \mathrm{MeV}/c.$$

Conclusions and Next Actions

Conclusions

- The pipeline reconstructs proton and neutron 3D momentum from synthetic input.
- Neutron TOF reconstruction gives moderate P_x bias and finite spread.
- Proton Minuit currently shows large positive ΔP_x bias in this sample.

Immediate next actions

- Diagnose proton Minuit bias with event-level (P_x^{true}, P_x^{reco}) outliers.
- Re-check coordinate/sign conventions between beam truth and reconstruction frame.
- Add per-bin P_x^{true} calibration/validation before production usage.